TANet: Thin Cloud-Aware Network for Cloud Detection in Optical Remote Sensing Image

Xinyi Xu, Wei He, *Senior Member, IEEE*, Yu Xia, Hongyan Zhang, *Senior Member, IEEE*, Yuwei Wu, Zhen Jiang, and Ting Hu

***Abstract*— Accurate cloud detection is essential for subsequent optical remote sensing imagery processing. Although many deep learning (DL)-based cloud detection methods have been proposed, the accurate detection of thin clouds still remains a challenge. To solve this issue, this article introduces a thin cloud-aware network (TANet). TANet tackles the problem from the aspects of color, texture, spatial distribution, and feature by constructing unique strategies to enhance the sensitivity of the network to thin cloud regions, thereby improving the overall accuracy of cloud detection. On the one hand, the TANet utilizes a color prior guidance module (CPGM) to incorporate robust dehaze priors, guiding the network to pay more attention to thin cloud areas. On the other hand, the global information aggregation module (GIAM) is employed to deeply extract long-distance dependence between pixels, mining potential correlations between thin and thick clouds, and addressing the challenge of recognizing thin clouds in a local perspective. In addition, we construct a plug-and-play cloud feature difference (CFD) loss, encouraging the network to learn more distinctive features between pixels of thin clouds and cloud-free regions, thereby strengthening the network's ability to distinguish highly similar samples of different classes. The experimental results substantiate that our proposed method attains the lowest omission error and the highest detection accuracy. This affirms the superior capability of TANet in thin cloud detection, thereby yielding more dependable cloud detection results.**

***Index Terms*— Cloud detection, deep learning (DL), dehaze prior, loss function, nonlocal, remote sensing image.**

I. Introduction

In RECENT years, optical remote sensing satellite images have been widely used for meteorological forecasting [1], [2], land use analysis [3], [4], and environmental monitoring [5], [6]. However, most optical remote sensing images are easily contaminated by clouds, and according to the report of the Intergovernmental Panel on Climate Change, nearly 66% of the Earth's surface is covered by clouds [7]. The presence of clouds may obstruct objects of interest and bias the images, leading to serious degradation of the performance of data applications. Therefore, fast and accurate detection of cloud-containing regions in remote sensing images is crucial for subsequent data preprocessing.

In optical remote sensing images, clouds can be classified as either thin or thick based on their thickness. The objective of cloud detection is to accurately identify regions in remote sensing images that contain clouds. In general, the existing cloud detection methods can be divided into threshold-based methods and machine learning methods [8]. Threshold-based methods are commonly used to detect clouds via multispectral bands by constructing decision rules based on the spectral differences between cloud pixels and clear pixels. Zhu and Woodcock [9] proposed the Fmask algorithm, which utilizes the unique physical properties of clouds to manifest features in different spectral bands of optical images. By establishing several spectral threshold conditions, Fmask ultimately achieves the automation of cloud detection. Wang et al. [10] devised a method named STI-ACSS, which was based on the HOT transform, introduced a novel adaptive thresholding technique to generate the initial cloud mask, and processed it in a morphological way to get the final result. Zhai et al. [11] designed the cloud index (CI) and the cloud shadow index (CSI) based on the physical reflective characteristics of clouds to indicate potential clouds and cloud shadows. These techniques are easy to implement and commonly used but still have some shortcomings. On the one hand, the selection of physical rules and optimal parameters in these algorithms relies on empirical determination and parameter sensitivity analysis, thereby introducing difficulties in achieving parameter adaption and global optimality and resulting in different degrees of cloud cover estimation bias in cloud masks [12]. On the other hand, such approaches rely solely on the spectral information of the imagery, leading to limited feature extraction. As a result, their performance is suboptimal in complex scenes such as snowy environments [13]. Moreover, the utilization of multiband data places demands on the number of satellite bands available, thereby restricting its applicability and generalization.

With the vigorous development of artificial intelligence algorithms, some scholars have employed labeled data and manually designed features to develop a series of machine

Received 24 January 2024; revised 29 June 2024 and 25 October 2024; accepted 30 January 2025. Date of publication 11 February 2025; date of current version 25 February 2025. This work was supported in part by the National Key Research and Development Program of China under Grant 2022YFB3903605, and in part by the National Natural Science Foundation of China under Grant 42071322 and Grant 42201377. *(Corresponding author: Ting Hu.)*

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Digital Object Identifier 10.1109/TGRS.2025.3540778

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learning-based methods. Hégarat-Mascle and André [14] utilized Markov random fields for automatic cloud and cloud shadow detection in high-resolution optical images based on three spatial features of clouds and cloud shadows. Li et al. [15] introduced the support vector machine [16] into cloud detection tasks, utilizing brightness and texture features of optical remote sensing images to train a support classifier for cloud detection. Wei et al. [17] chose the stratified sampling method to preselect cloudy and clear-sky pixels to form a prior-pixel database and then used typical spectral indices as inputs to the random forest model [18] for training and establishing a preliminary cloud detection model. These algorithms introduce statistical and texture features of images as the basis for classification, improving the accuracy of cloud detection to some extent. However, due to the fact that the quality of features still depends on manual design, and considering the highly complex environment in remote sensing images, the practical applicability of these methods is limited [19].

As a branch of machine learning, deep learning (DL) using neural networks can automatically extract multilevel and nonlinear features with end-to-end learning [20], which has shown excellent performance in various image tasks [21], [22], [23]. More and more scholars are focusing on using neural networks to solve the processing and analysis tasks of remote sensing images [24], [25], [26], [27], including cloud detection, and are gradually leading the development of this field. Xie et al. [28] used the simple linear iterative clustering (SLIC) [29] method to divide images into good-quality superpixels, then trained a deep convolutional neural network to classify them, and, finally, reassembled to obtain a complete cloud mask. Jeppesen et al. [30] designed a DL model for cloud detection based on the Unet [31] architecture, which pioneered the idea of image semantic segmentation based on fully convolutional neural networks in the field of cloud detection. Shao et al. [32] introduced a convolutional neural network based on multiscale feature convolution, using ten bands of Landsat 8 to achieve cloud detection in remote sensing images. Guo et al. [33] introduced an adaptive feature fusion model (AFFM) and high-level semantic information guidance flows (HSIGFs) to form CDnetV2, which increases the accuracy of cloud detection by combining multilevel feature maps and compensating for position information in semantic features. Li et al. [34] proposed a lightweight cloud detection network CD-FM3SF that uses three novel components in the encoding process and uses deep separable convolution (DSC) in the decoder to fuse the same level of features in the encoder and the decoder. Wu et al. [35] innovatively proposed a cloud detection boundary network composed of two networks: a scalable-boundary network and a differentiable-boundary network, which aims to solve the issues of scale insensitivity and boundary fuzziness in DL-based cloud detection methods. Chen et al. [36] proposed a CI-driven spectral–spatial-context attention network (SSCA-net), which equipped the spectral–spatial-context aware module and the spectral–spatial-context information aggregation module to improve the accuracy of cloud detection in high-brightness scenes.

In conclusion, DL algorithms have shown significant advantages in cloud detection, especially when enhanced with multiband data and various multiscale modules. However, existing DL models still exhibit certain shortcomings in cloud detection. The results of the MFCNN experiments [32] indicate that thin clouds are significantly underdetected compared to thick clouds in all evaluation metrics, suggesting that their treatment of thin clouds lacks depth. Similarly, Li et al. [34] found that the limited number of thin cloud samples in the experimental dataset hindered the CD-FM3SF model's ability to accurately detect thin clouds, despite its balanced model size and accuracy. Li et al. [37] also noted that the small size, sparse distribution, and high transparency of thin clouds create similarities to noncloud background regions, making it easy for existing methods to fail in accurately processing thin clouds.

DL-based methods generally consider cloud detection as a specialized semantic segmentation task based on remote sensing datasets. They typically utilize fully convolutional neural networks [38], which employ data-driven learning to extract features, enabling the development of effective cloud detection models that yield promising results on various datasets. Most current methods primarily focus on the overall spatial scale characteristics of clouds, without carefully considering the feature differences between different cloud regions due to variations in cloud thickness. Clouds can be classified as thick or thin based on their optical thickness [39]. Thick clouds typically obscure the surface entirely and display prominent spectral, spatial, and textural features. In contrast to thick clouds, which are easily identifiable, thin cloud regions appear translucent in optical remote sensing images. These regions contain not only cloud information but also details of the underlying surface, including color and texture [40], as illustrated in Fig. 1(a) and (b). Furthermore, as shown in Fig. 1(c) and (d), thin clouds are often found at the edges of thick clouds or occur independently in isolated patches. As a result, during the detection process, thin clouds are susceptible to being overlooked [41]. However, few existing studies have considered these factors, leading to imprecise detection of thin clouds in the results [42]. On the one hand, this significant issue of thin cloud confusion hinders overall detection accuracy, which consequently impacts data processing in subsequent applications [43]. On the other hand, areas obscured by thick clouds are often deemed invalid information, while regions with thin clouds have the potential for information recovery [44]. Therefore, accurate identification of thin cloud regions is a crucial step in cloud removal methods [45]. For these reasons, this issue requires urgent further research and resolution to enhance the overall accuracy of cloud detection and provide high-quality data support for subsequent applications.

In order to address the above issues, this article introduces a thin cloud-aware cloud detection network, abbreviated as TANet. This network, based on an encoder–decoder structure, strengthens the learning of features conducive to thin cloud detection in various aspects, including color, texture, spatial distribution, and feature. Specifically, considering the similarity in color and texture between thin clouds and underlying

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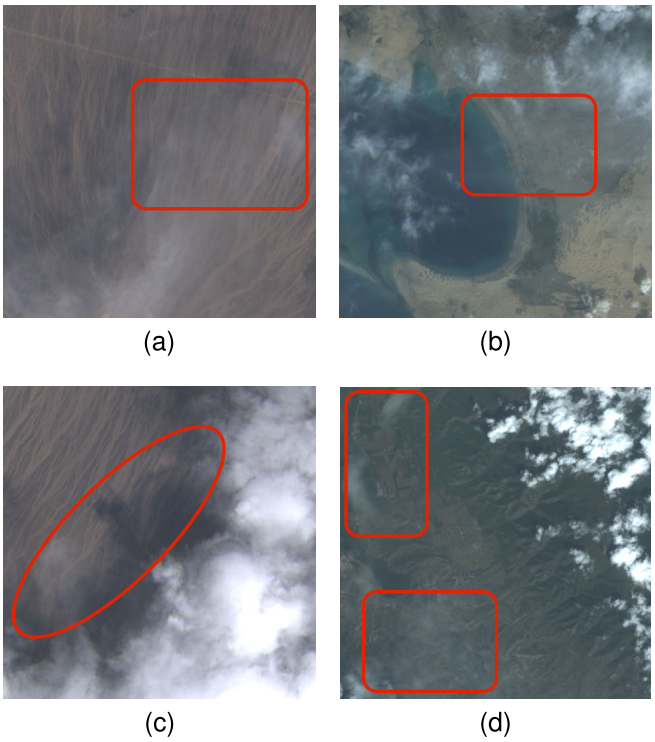


Fig. 1. (a) and (b) Semitransparent, thin clouds do not fully cover the subsurface, displaying similar color and texture properties. (c) Some thin clouds at the junction of thick and cloudless areas. (d) Some thin clouds that are isolated from the thicker ones.

surfaces in images, as well as the spatial distribution characteristics of thin clouds, we design a color prior guidance module (CPGM) and a global information aggregation module (GIAM). The CPGM introduces classic robust dehaze priors to generate color prior guidance maps, which highlight thin cloud regions and suppress irrelevant areas, thereby making the network more focused on the thin cloud regions. The GIAM establishes long-distance dependencies between pixels, exploring potential connections between thin and thick cloud pixels. This allows the module to obtain enhanced feature representations, mitigating interference from local textures on the underlying surface. In addition, to further force the network to learn beneficial detailed features, we introduce a cloud feature difference (CFD) loss that can be integrated into any existing cloud detection network. Under the supervised training of this loss, the network is compeled to extract the similarities and differences in features between thin and thick clouds and the underlying surface. This process enables the network to learn distinctive features, thereby reinforcing its capacity to discriminate between highly similar samples belonging to different classes. The integration of the above strategies forms our method. In summary, our contribution includes the following three parts.

1) We devise a CPGM based on a lightweight convolutional encoder and a GIAM with a nonlocal structure. The CPGM incorporates robust dehaze priors, emphasizing the relevant thin cloud regions and suppressing irrelevant areas. The GIAM deeply extracts the global features and potential correlations between pixels, resisting the adverse effects of underlying surface local textures.
2) We propose a plug-and-play CFD loss, which aims to encourage the network to learn differentiated features between thin clouds and easily confused underlying surfaces, thereby increasing the network's capacity to perceive thin clouds.
3) We conducted extensive experiments on cloud detection datasets from Landsat-8 and Sentinel-2. The results indicate that the TANet can not only detect thin clouds more accurately but also obtain more reliable results with better robustness in various subsurfaces and complex cloud distribution scenarios. The novel CFD loss can guide the training of any existing cloud detection network effectively so that the corresponding network can obtain stronger thin cloud detection capability to achieve better overall prediction performance.

This article is an extension based on the previous conference article [46]. In contrast to [46], we extend our work as follows.

1) In [46], we have conducted a general analysis of the confusion detection caused by thin clouds and proposed the CFD loss for the first time. In this article, we further comprehensively investigate the essential features of thin clouds and refine our solution.
2) Different from previous work that assists existing networks to achieve thin cloud detection through indirect way, this article proposes a novel cloud detection network that provides a direct solution.
3) We evaluate the effectiveness of the CFD loss with additional experiments. By performing ablation learning on multiple advanced cloud detection networks, we have provided further support for the validity of our approach.

II. Method

Thin clouds, due to their semitransparent nature, are prone to being overlooked during the detection process, limiting the overall detection effectiveness. To address the difficult challenge of thin cloud detection, this article introduces a thin cloud-aware network (TANet). In this section, we first provide a comprehensive overview of our proposed architecture and strategies and then describe each module in detail.

A. Overall Architecture

The overall architecture of the proposed TANet is illustrated in Fig. 2. We adopt the RS-Net [30] as the basic framework for cloud detection. The network as a whole uses an encoder–decoder structure, striking a balance between lightweight structure and high performance. It has been demonstrated to achieve accurate cloud detection under conditions with fewer spectral bands.

Building upon the backbone network, we integrated strategies based on color, texture, spatial distribution, and feature to strengthen the network's capacity for modeling thin cloud characteristics. This amalgamation ultimately forms the TANet. Specifically, we utilize RGB input images $\mathbf{I} \in \mathbb{R}^{3\times H\times W}$ to generate the corresponding dark channel prior $\mathbf{P_{DCP}} \in \mathbb{R}^{1\times H\times W}$ and color attenuation prior (CAP) $\mathbf{P_{CAP}} \in \mathbb{R}^{1\times H\times W}$.

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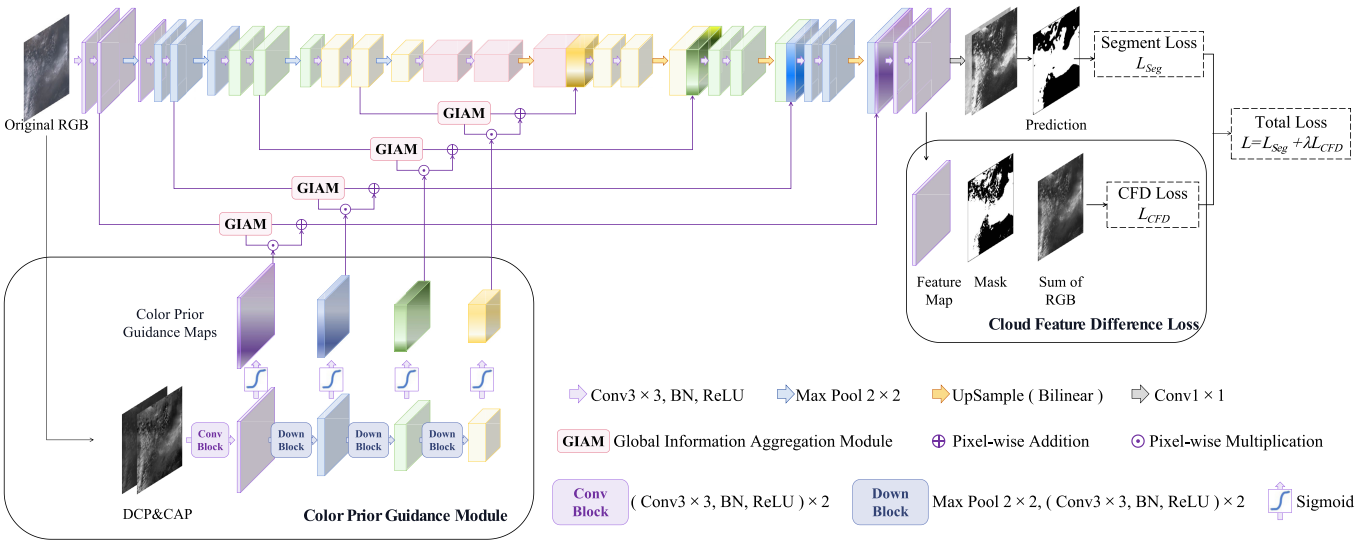


Fig. 2. Framework of TANet. The CPGM utilizes a lightweight and independent encoder structure combined with a sigmoid to generate guidance maps focusing on thin cloud areas. The GIAM is a nonlocal module designed to capture long-range dependencies in features at each layer. The CFD loss is a plug-and-play loss function that can guide the network with classical segmentation loss.

Through an independent CPGM, we extract color prior guidance maps $\mathbf{M} = \{m_l\}_{l=1}^{4}$ at different scales

$$m_l = \text{Sigmoid}(m_l') \tag{1}$$

$$m_l' = \begin{cases} \text{CBR}_l^C(\text{Concat}(\mathbf{P_{DCP}}, \mathbf{P_{CAP}})), & \text{if } l = 1, \\ \text{CBR}_l^C(\text{Maxpool}(m_{l-1}')) & \text{otherwise} \end{cases} \tag{2}$$

where $\text{CBR}_l^C(\cdot)$ represents the convolutional block containing two convolution operations with batch normalization [47] and ReLU activation function [48] in the lth layer of the CPGM. The final color prior guidance map m_l is obtained by applying the sigmoid function to the output of each layer in CPGM.

For a given RGB input image $\mathbf{I}$, the encoder progressively extracts features $\mathbf{F} = \{f_l\}_{l=1}^{5}$ from shallow to deep layers through convolution and pooling operations

$$f_l = \begin{cases} \text{CBR}_l^e(\mathbf{I}), & \text{if } l = 1, \\ \text{CBR}_l^e(\text{Maxpool}(f_{l-1})) & \text{otherwise} \end{cases} \tag{3}$$

where $\text{CBR}_l^e(\cdot)$ denotes the convolution block with the same structure as $\text{CBR}_l^C(\cdot)$ in the lth layer in encoder and f_l is the features obtained by each layer. We employ a GIAM to process f_l

$$f_l^G = \text{GIAM}(f_l) \tag{4}$$

where f_l^G represents the features obtained with global context-dependent information. Subsequently, these features are multiplied pixelwise with the corresponding scale of color prior guidance maps and perform residual concatenation to obtain the final feature maps as

$$f_l' = f_l^G + f_l^G \cdot m_l. \tag{5}$$

These feature maps are focused on thin clouds and encompass global information. To avoid interfering with the encoder's extraction of original features, we do not use the above strategies at each layer of the encoding process. Instead, we exclusively process the feature maps from the first four layers to obtain results, which were then used as features fused with the decoder through skip connections. The decoder, through upsampling and convolution, restores the high-level semantic feature maps $\mathbf{F^d} = \{f_l^d\}_{l=1}^{4}$ obtained by the encoder to the original image resolution, as formulated by the following equation:

$$f_l^d = \begin{cases} \text{CBR}_l^d(\text{Concat}(f_l', \text{Up}(f_5))) & \text{if } l = 4, \\ \text{CBR}_l^d(\text{Concat}(f_l', \text{Up}(f_{l+1}^d))) & \text{otherwise} \end{cases} \tag{6}$$

where $\text{CBR}_l^d(\cdot)$ denotes the convolution block for each layer in the decoder, $\text{Up}(\cdot)$ stands for the upsampling operation, bilinear interpolation algorithm is used here, and f_l^d is the features interpreted by each layer. The final predicted result $\mathbf{Y}$ is

$$\mathbf{Y} = \text{Softmax}(\text{OutConv}(f_1^d)). \tag{7}$$

$\text{OutConv}(\cdot)$ is a convolution followed by batch normalization and ReLU activation function. Finally, to encourage the network to model the similarity and differentiation features among different samples, we use the proposed CFD loss combined with segmentation loss to guide the network's training.

B. Color Prior Guidance Module

The fundamental principle of cloud detection is utilizing effective features to distinguish between the subsurface and the cloud. However, the semitransparency of thin clouds results in a lack of distinct cloud features in visible imagery, making them similar to neighboring underlying surface regions, as shown in Fig. 1(a) and (b). In order to make the network more attentive to the regions of interest, we plan to use robust and reliable prior information to suppress the interference of cloud-free areas and highlight thin cloud regions. Through analysis, we find that unlike the bright thick clouds, the region of thin clouds appears blurry and dark. The formation mechanism of clouds and fog is highly similar, and their

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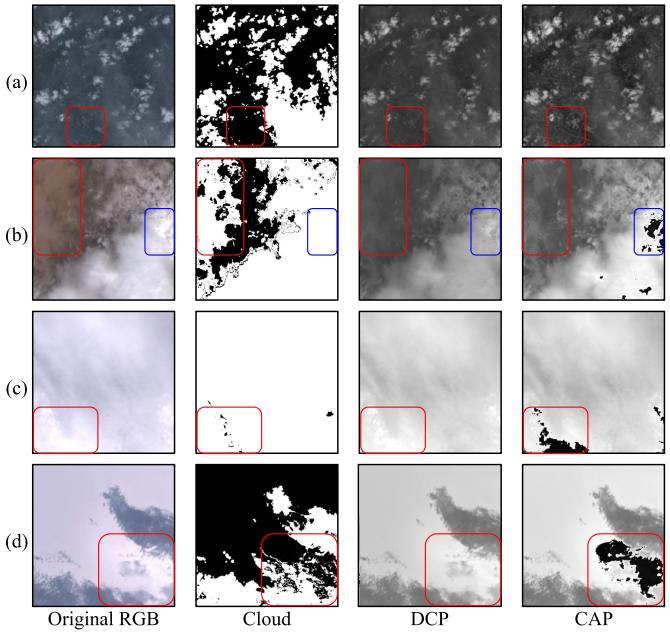


Fig. 3. Examples of DCP and CAP corresponding to visible remote sensing imagery with different subsurface and cloud cover. (a) Grass/crops. (b) Shrubland. (c) Bright thick clouds on the underlying surface of ice and snow. (d) Clouds also have the potential to reduce the brightness of the ice/snow subsurface on the image, with exposed cloud-free areas being relatively brighter.

imaging characteristics in optical imagery are also comparable. In studies on images with fog, some color-based priors have been proposed, proving effective in highlighting foggy areas with strong robustness. This inspired us to design the color prior guidance module, which incorporates classical dark channel prior [49] and CAP [50] to guide the network to focus more attention on the regions of interest, particularly thin cloud areas.

The dark channel prior (DCP) observes that in fog-free images, nonsky and nonpure white regions typically have pixel values close to zero in at least one channel among the RGB three channels. Conversely, in images with fog, the darkest channel's values of foggy pixels are much greater than zero. Mathematically, given an RGB image $I(x)$ and its corresponding dark channel $J(x)$, where x represents pixel coordinates and $J(x)$ is defined as the minimum value of $I(x)$ within a local window, the dark channel prior can be expressed as

$$J(x) = \min_{y\in\Omega(x)}(\min_c I^c(y)). \tag{8}$$

When applied to RGB remote sensing images, the generated DCP maintains high brightness in thin cloud regions while presenting darker colors in noncloud underlying surface regions, and the texture features are also weakened, as shown in the red part of Fig. 3(a) and (b).

The CAP is based on the finding that the concentration of fog is directly proportional to the difference between brightness and saturation. Through extensive samples, the relationship between depth of field and the difference between brightness and saturation is established through the following formula:

$$d(x) = \theta_0 + \theta_1 v(x) + \theta_2 s(x) + \epsilon(x) \tag{9}$$

where x is the position within the image, d is the scene depth, v is the luminance component of the fogged image, and s is the saturation component. θ_0, θ_1, and θ_2 are the linear coefficients, and $\epsilon(x)$ is a random error that follows a Gaussian distribution with a mean of 0 and a variance of σ^2. According to the setting $\theta_0 = 0.121779, \theta_1 = 0.959710, \theta_2 = -0.780245$, and $\sigma = 0.041337$ in the original article, applying this linear model to image, the corresponding prior can be obtained. In some areas, the light reflection is so strong that the CAP cannot correctly distinguish between the true color and the attenuation effect in these areas, so the values calculated in these areas are abnormally low, resulting in black patches. As can be seen in the red part of Fig. 3(b), when applied to cloud images over general underlying surfaces, the CAP makes the previously transparent and inconspicuous thin cloud areas more distinguishable. It should be noted that the thick cloud areas in the optical remote sensing image are too bright, resulting in dark patches in some areas, such as the blue part of Fig. 3(b). Special underlying surfaces, such as ice and snow, on the one hand, due to the overlying of clouds, and some ultra-bright thick clouds areas on the CAP are displayed as black blocks, as shown in Fig. 3(c), which is similar to most underlying surfaces. However, on the other hand, there are some bare snow and ice without clouds that may be brighter than those covered with clouds. Instead, the cloud coverage makes the brightness of the cloudy area relatively low, causing black blocks to appear in the ultrabright area without clouds [see Fig. 3(d)]. This feature presented by CAP can generate auxiliary information for the extremely bright surface and provide a prior guidance support for the challenging underlying surface. Therefore, our proposed color prior guidance module is used to process these two prior information, generating multiscale color prior guidance maps.

To be specific, as Fig. 4 shows, the CPGM employs a lightweight convolutional encoder to process the concatenated two prior maps. It uses convolution and pooling operations to extract color prior features at different scales. Subsequently, an activation function is applied to generate corresponding attention weights, forming the final color prior guidance maps. The color prior guidance maps, generated at various scales, are subsequently utilized to process features obtained by the encoder at each layer. This process ensures that the network allocates greater attention and resources to thin cloud regions and challenging bright surfaces during training, thereby contributing to the enhancement of precise thin cloud detection.

C. Global Information Aggregation Module

In optical remote sensing images, thin clouds cannot completely obscure the underlying surface, revealing partial ground texture features, making it challenging to distinguish them from a local perspective. Moreover, thin clouds often exist at the boundary of cloud regions or even isolated from thick cloud areas, rather than being adjacent to thick clouds, as illustrated in Fig. 1(c) and (d). Mainstream cloud detection

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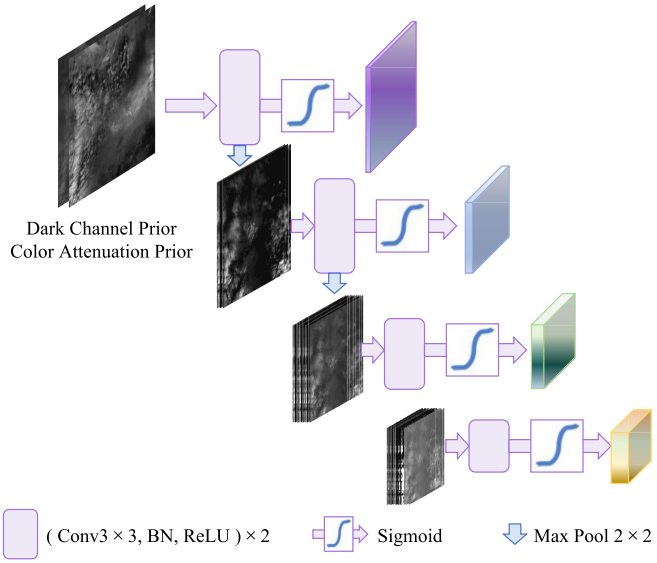


Fig. 4. CPGM. Since the strategy is used only for skip connection in the baseline network, the CPGM module only generates color prior guidance maps for the four layers corresponding to the scales.

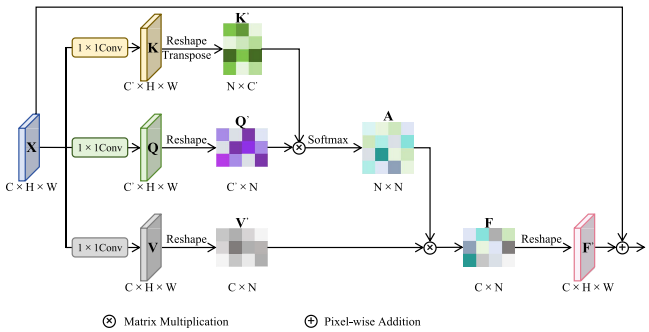


Fig. 5. Details of GIAM. The number of channels of K and Q obtained by 1×1 convolution $C' = (1/8)C$; the final output obtained captures the rich potential connections between pixels and global contextual information.

methods, mostly based on convolutional layers with limited receptive fields [51], tend to extract local texture features and cannot model the long-range spatial correlations between distant pixels [52]. Starting from this fact, we construct a GIAM based on a nonlocal structure [53], with details shown in Fig. 5. This module effectively explores the potential feature connections between thin and thick cloud pixels, thereby mitigating the adverse effects caused by local subsurface texture on thin cloud detection.

Specifically, for the input feature $\mathbf{X} \in \mathbb{R}^{C\times H\times W}$, three independent 1×1 convolutions are employed for linear mapping to obtain key ($\mathbf{K} \in \mathbb{R}^{C'\times H\times W}$), query ($\mathbf{Q} \in \mathbb{R}^{C'\times H\times W}$), and value ($\mathbf{V} \in \mathbb{R}^{C\times H\times W}$)

$$\mathbf{K} = \mathrm{Conv}^{K}_{1\times 1}(\mathbf{X}) \tag{10}$$

$$\mathbf{Q} = \mathrm{Conv}^{Q}_{1\times 1}(\mathbf{X}) \tag{11}$$

$$\mathbf{V} = \mathrm{Conv}^{V}_{1\times 1}(\mathbf{X}) \tag{12}$$

where $\mathrm{Conv}_{1\times 1}$ represents a convolutional layer with a kernel size of 1. The channels of $\mathbf{K}$ and $\mathbf{Q}$ are compressed to one-eighth of their original size. Through the reshape operation, $\mathbf{K}$, $\mathbf{Q}$, and $\mathbf{V}$ become $\mathbf{K}' \in \mathbb{R}^{N\times C'}$, $\mathbf{Q}' \in \mathbb{R}^{C'\times N}$, and $\mathbf{V}' \in \mathbb{R}^{C\times N}$, respectively, where $N = H \times W$. The rough attention map is obtained by matrix multiplication, and then, the softmax function is applied to normalize it, generating the final weight map $\mathbf{A} \in \mathbb{R}^{N\times N}$ containing the relationship between each pixel and all other points

$$\mathbf{A} = \mathrm{Softmax}(\mathbf{Q}'\mathbf{K}'). \tag{13}$$

Finally, the weight map is multiplied with V', and a residual operation is performed with the original input feature map to get the output result $\mathbf{X}' \in \mathbb{R}^{C\times H\times W}$

$$\mathbf{X}' = \mathrm{Reshape}(\mathbf{V}'\mathbf{A}) + \mathbf{X}. \tag{14}$$

In the above way, the global information of the feature map and the long-distance dependence between pixels can be modeled. Even when thin cloud pixels are distant from thick cloud pixels, the GIAM can capture their potential connection and provide global information to resist the confusion caused by the local perspective.

D. CFD Loss

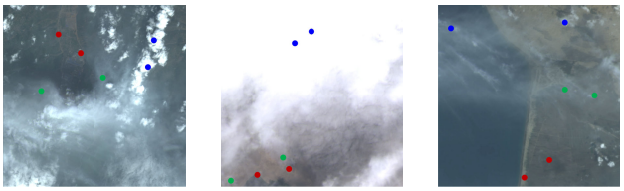
The cloud detection network learns features autonomously, driven by data, and applies these features for pixel classification. Ideally, we expect the learned features of samples from different classes to exhibit sufficient differences among each other to ensure the network's accuracy in the cloud detection task. However, within cloud regions, the features of thick clouds are prominent and distinctive. In contrast, the features of thin clouds, while somewhat similar to thick clouds, are relatively concealed and more akin to the features of noncloud underlying surfaces. This makes it a challenging task for the network to autonomously achieve accurate identification of thin clouds. To reinforce the neural network's capability to differentiate extremely similar samples that are not of the same class, we design a CFD loss inspired by triplet loss [54]. The CFD loss calculates the absolute and relative feature distances between thin cloud samples, thick cloud samples, and clear samples. By optimizing the CFD loss, it is possible to narrow the feature distance between thin cloud and thick cloud samples while increasing the distance between thin cloud and underlying surface samples. Through this explicit and strong supervision method, the network is compeled to learn differentiating features between thin clouds and cloud-free regions, thereby enhancing its ability to detect thin clouds.

For CFD loss computation, we introduce the concept of a cloud triplet comprising cloud positive (CP), cloud anchor (CA), and cloud negative (CN) samples. As Fig. 6 shows, CP and CA samples represent thick and semitransparent thin cloud pixels, respectively, while CN samples represent noncloud subsurface pixels. In detail, for a cloud triplet, the CFD loss is calculated as follows:

$$L_{\mathrm{CFD}} = \max(d(\mathrm{CA}, \mathrm{CP}) - d(\mathrm{CA}, \mathrm{CN}) + \mathrm{margin}, 0) \tag{15}$$

where CA, CP, and CN represent the corresponding sample points and $d(\cdot, \cdot)$ calculates the feature distance. margin is a hyperparameter greater than 0. To minimize the CFD loss, it is essential for $d(\mathrm{CA}, \mathrm{CP}) - d(\mathrm{CA}, \mathrm{CN})$ to approach margin. Specifically, this requires $d(\mathrm{CA}, CP) < d(\mathrm{CA}, \mathrm{CN})$ with the

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(a) (b) (c)

Fig. 6. Examples of Triplets of different categories. (a) Simple triplets. (b) Hard triplets. (c) Semihard triplets. Blue points: CP samples. Green points: CA samples. Red points: CN samples.

difference between the two distances being close to margin. The CFD loss provides strong explicit supervision, guiding the network to achieve the state where $d(\text{CA}, \text{CP}) < d(\text{CA}, \text{CN})$ in its mapped feature space.

From the above expression, we can observe that to accurately calculate the CFD loss, it is essential to select corresponding cloud-triplet sample points, compute distances in a specific feature space, and ultimately obtain the loss value. Next, we describe in detail the two key steps and the setting of margin, respectively.

1) Adaptive Cloud-Triplet Selection: Based on the proximity of the features of the selected CA points and CN samples, the combined cloud triplets can be classified into three categories as follows.

a) Simple triplets: CA samples have a certain degree of transparency but can still obscure most of the underlying surface information. In this case, $d(\text{CA}, \text{CP})$ is naturally small, while the distance between $d(\text{CA}, \text{CN})$ is large, requiring no optimization.

b) Hard triplets: CA samples are thin clouds with high transparency, CP samples are thick clouds with significant cloud features, and CP samples are noncloud regions with features similar to clouds. In this case, due to the near-transparency of thin clouds, which includes some underlying surface information, the spectral and textural characteristics resemble ground objects. $d(\text{CA}, \text{CN})$ is smaller than $d(\text{CA}, \text{CP})$ resulting in a large loss that needs optimization.

c) Semihard triplets: CA samples are thin clouds, but CP samples have insufficient optical thickness, leading to less prominent cloud features, or the underlying surface does not have easily confused samples. In these cases, $d(\text{CA}, \text{CN})$ may be similar to $d(\text{CA}, \text{CP})$ but does not satisfy the condition where the difference is greater than margin. Loss is still present but smaller than that of hard triplets, and optimization is needed. By analyzing these three types of cloud triplets, it is clear that it is necessary to select hard or semihard triplets as much as possible, identifying the most easily confused thin cloud and clear pixels rather than simply selecting samples from different categories. Relying on manual visual selection would be a costly task. Therefore, we design an efficient adaptive cloud-triplet selection strategy considering diverse reflectance properties across subsurfaces and clouds. The selection process only requires the sum of reflectance in the R, G, and B channels, along with real cloud mask data. Due to the semitransparency of thin clouds, their reflectance characteristics in the visible light spectrum are more similar to the underlying surface. We select m pixels with the lowest sum of reflectance in cloud areas as CA samples, roughly expected to be susceptible to confusion with the subsurface. For CP samples representing thick clouds with high reflectance, we choose n pixels with the highest sum of reflectance in cloud areas. The k pixels with the highest sum of reflectance in noncloud pixels are considered CN samples, assuming that these k pixels are most likely to be confused with thin cloud pixels. Given that experimental data often undergo cropping, leading to blocks with few or no cloud or clear pixels, we opt to select a limited number of sample points. This approach utilizes as much as possible of each image patch that has clouds, expediting the acquisition of the required cloud triplet and reducing the computational load.

2) Feature Selection: Based on the selected cloud triplet, it is necessary to choose a specific feature space for loss calculation. We consider that the lower layers respond to primitive image features such as edges, corners, and shared common patterns [55]. In this space, thin clouds are already very similar to the underlying surface, making it difficult to achieve feature differentiation. Deeper outputs encode high-level semantic knowledge [56] but exhibit lower resolution. Directly sampling deep features to the original resolution for calculating feature distances may introduce significant errors. The features obtained after decoder processing not only integrate shallow and deep features but also include color prior guidance information and global dependence information. Simultaneously, these features maintain a resolution closer to the original size. Therefore, in the calculation process of the CFD loss, we select the output features before the last convolutional layer as the basis for distance calculation.

3) Setting of Margin: margin is a constant greater than 0. Its purpose is to avoid shortcuts in the model. This is because when margin $= 0$, the CFD loss becomes

$$L_{\text{CFD}} = \max(d(\text{CA}, \text{CP}) - d(\text{CA}, \text{CN}), 0). \tag{16}$$

At this point, the model may learn the situation where $d(\text{CA}, \text{CP}) = d(\text{CA}, \text{CN})$, indicating equivalent distances between CA samples and CP samples, as well as CA samples and CN samples. This leads to training failure, rendering the model unable to correctly classify CA samples. Setting a margin can force the model to train correctly. A larger margin enhances the model's ability to discriminate between very similar samples of different classes, but setting a relatively large value initially may increase the difficulty of model training, potentially preventing network convergence. Therefore, we statistically calculate the initialized value of $d(\text{CA}, \text{CP}) - d(\text{CA}, \text{CN})$ and set a suitable small value. Through experiments, we gradually adjust margin, ensuring network convergence while achieving optimal performance.

Fig. 7 illustrates the main calculation steps for CFD loss. The CFD loss calculated through the above steps can be integrated into any existing cloud detection network. When combining the CFD loss with classical segmentation loss to guide network training, the total loss to be optimized is expressed as the following equation:

$$L = L_{\text{seg}} + \lambda L_{\text{CFD}} \tag{17}$$

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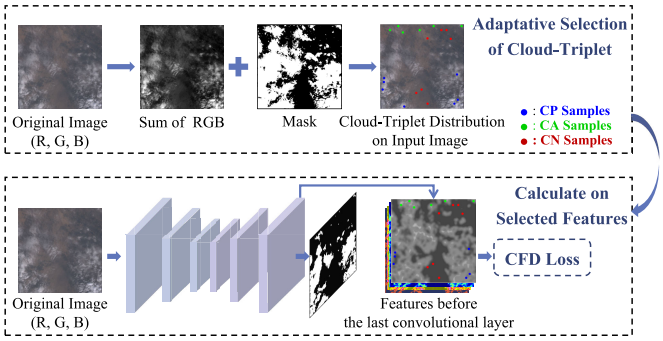


Fig. 7. Calculation process of the CFD loss. Blue points: CP samples. Green points: CA samples. Red points: CN samples.

where λ represents the weight of CFD loss, which needs to be adjusted manually according to different situations. We incorporate CFD loss into our overall approach. Under the supervision of the above loss, TANet's ability to extract characteristics of differentiation is significantly increased, leading to a more sensitive perception of thin clouds, which, in turn, leads to superior cloud detection results.

III. Experiment

A. Experimental Settings

1) Data Preparation: The dataset used in the experiment includes Landsat-8 cloud cover assessment validation data [57] and Sentinel-2 Cloud Mask Catalogue [58].

The Landsat-8 cloud cover assessment validation data includes 96 Landsat-8 images. These images were randomly selected from around the world using a stratified sampling method based on eight biomes. Each image has a manually annotated cloud mask. For each type of underlying surface, we selected four images as training data and two images as test data to ensure independence. The cloud coverage level of all the selected images is medium to make the number of positive and negative samples as balanced as possible. The cloud masks for this dataset are labeled with four categories: thick cloud, thin cloud, clear, and cloud shadow.

The Sentinel-2 Cloud Mask Catalogue dataset [58] contains 513 subscenes, each with dimensions of 1022 $\times$ 1022 pixels, annotated with labels for clear sky and cloud only. The data, with a resolution of 20 m, are randomly sampled from the 2018 Level-1C Sentinel-2 archive. For the experiments, 410 images (approximately 80%) were randomly selected from the 513 images to form the training dataset, while the remaining images were used as the base data for the validation dataset.

Based on the original images from two datasets, cropping was performed with a size of 512 $\times$ 512 and a 100-pixel overlap to obtain the reconstructed data for the experiment. To validate the low dependence of our method on multiband data and its generalizability, we used only the RGB three-band data as input.

2) Comparison Methods: The methods used for comparison include classical semantic segmentation algorithms, Unet [31] and DeeplabV3 [59], and state-of-the-art cloud detection methods: MFCNN [32], CDnet [60], CDnetV2 [33], MSCFF [61], and Boundary Nets [35]. A brief overview of MFCNN, CDnetV2, and Boundary Nets can be found in Section I. CDnet is an encoder–decoder structured network with a feature pyramid module (FPM) and a boundary refinement (BR) module. MSCFF is based on a symmetric encoder–decoder structure, using a multiscale feature fusion module to combine features from different scales for obtaining results.

All networks were trained on our reconstructed dataset, and relevant training strategies were set according to the recommendations in the original articles.

3) Implemented Details: TANet is implemented on the PyTorch DL framework. The optimizer used in the experiment is Adam [62], with an initial learning rate of 0.001 and a weight decay of 5e-4. StepLR is employed as the learning rate adjustment strategy, where the step size is set to 15 and gamma is set to 0.5. The batch size is configured as 12. The CFD loss weight λ is set to 0.001, the margin is set to 0.25, and the number of sample points for the three kinds of cloud triplet is set $n = k = m = 10$. The encoder obtains feature channels with numbers of 64, 128, 256, 512, and 512 sequentially from shallow to deep layers. The decoder interprets the fused features from each skip-connected layer progressively, resulting in channel numbers decreasing from 512, 256, 128, to 64 as it transitions from deep to shallow layers. Finally, the result is obtained from the output convolution, and the cloud mask predicted by the output network consists of only two categories: cloud and clear.

4) Evaluation Metrics: It is important to note that the Landsat-8 cloud cover assessment validation data provide clear labels that distinguish thin and thick clouds, while the Sentinel-2 cloud mask catalog dataset does not make this distinction. In analyzing the experimental results for both datasets, we employed four common metrics: overall accuracy, kappa statistic, commission error, and omission error [63]. These metrics provide a comprehensive evaluation of the methods.

The overall accuracy fully reflects the prediction accuracy of the method. The kappa statistic represents the consistency between the network's prediction results and the true masks. The commission error represents the probability that the network misidentifies clear pixels as cloud pixels. The omission error represents the probability that the network incorrectly recognizes cloud pixels as clear pixels, which is a crucial indicator reflecting the missed detection situation of thin clouds. These metrics are computed based on the confusion matrix, and the specific calculations are given as follows:

$$\text{Overall Accurancy} = \frac{tp + tn}{tp + fp + tn + fn} \tag{18}$$

$$\text{Kappa} = \frac{p_0 - p_e}{1 - p_e} \tag{19a}$$

$$p_0 = \frac{tp + tn}{tp + fp + tn + fn} \tag{19b}$$

$$p_e = \frac{(tp + fn) \cdot (tp + fp) + (fp + tn) \cdot (fn + tn)}{(tp + fp + tn + fn)^2} \tag{19c}$$

$$\text{Commission Error} = \frac{fp}{fp + tn} \tag{20}$$

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TABLE I
3 × 2 CONFUSION MATRIX

True \ Predict	Cloud	Clear
Thick cloud	M_{11}	M_{12}
Thin cloud	M_{21}	M_{22}
Clear	M_{31}	M_{32}

$$\text{Omission Error} = \frac{fn}{tp + fn} \tag{21}$$

where tp denotes the number of pixels that are actually clouds and are correctly identified as clouds; fp denotes the number of pixels that are not actually clouds but are incorrectly identified as clouds; tn denotes the number of pixels that are not actually clouds and are not identified as clouds; and fn denotes the number of pixels that are actually clouds but are not identified as clouds.

In addition to the four metrics mentioned above, our objective was to fully utilize the detailed labeling information from the Landsat-8 cloud cover assessment validation data by constructing a 3 × 2 confusion matrix as in Table I, which outputs only cloud and clear predictions. This matrix enabled us to develop a novel class-specific omission error metric, facilitating a more intuitive analysis of thin cloud detection performance. The omission error of thick cloud is obtained as

$$\text{OE}_{\text{thick}} = \frac{M_{12}}{M_{11} + M_{12}}. \tag{22}$$

The omission error of thin cloud is obtained as

$$\text{OE}_{\text{thin}} = \frac{M_{22}}{M_{21} + M_{22}}. \tag{23}$$

By employing the class-specific omission error metric, we can further discern the sources of omission errors in the results, allowing a more targeted analysis of the challenges faced by different cloud categories during detection. It is important to highlight that this set of metrics is exclusively applied to the experimental results from the Landsat-8 cloud cover assessment validation data.

B. Experimental Results

1) Quantitative Results: Table II summarizes the quantitative results of different methods on the Landsat-8 cloud detection dataset. From the perspective of overall accuracy, the TANet outperforms other methods even without training using the additional CFD loss. After incorporating CFD loss for training assistance, the accuracy of our method is further improved. Meanwhile, the results reveal that most methods achieve relatively high omission errors, while commission errors tend to be lower and more consistent. This observation reinforces the previously discussed issue of omission in cloud detection. A closer examination of the class-specific omission error for thin and thick clouds shows that the omission error for thin clouds is significantly higher than that for thick clouds, indicating that misclassifications predominantly occur in thin cloud areas. Compared to other methods, TANet, trained exclusively with traditional segmentation loss, has already achieved the lowest overall omission error and the lowest omission error for thin clouds, highlighting its strong capability in thin cloud detection. With added training support from CFD loss, TANet's overall omission error further decreases by 7.22%, while the omission error for thick clouds also maintains an optimal level, resulting in the highest overall accuracy. These quantitative metrics clearly demonstrate that our method exhibits robust thin cloud detection capabilities, effectively addressing omission issues in cloud detection and yielding more accurate detection outcomes.

Table III summarizes the quantitative results of different methods on the Sentinel-2 Cloud Mask Catalogue dataset. Overall, the performance of various methods on the Sentinel-2 dataset is significantly better than on the Landsat-8 cloud detection dataset. Moreover, the differences in the accuracy metrics calculated for the prediction results of various methods are minimal. The reason is that the original data of the Sentinel-2 Cloud Mask Catalogue are compiled from randomly selected subscenes, some of which may have image overlaps or be located in the same region. This results in less independence between the reconstructed training and validation data, allowing the trained network to fit better on the validation data. In contrast, the original data of the same scene in the Landsat-8 cloud cover assessment validation data do not overlap, with greater differences between each other, resulting in a higher degree of independence between the reconstructed training and validation data and relatively poorer results.

Although the commission error and omission error on the Sentinel-2 dataset are generally low, there is still a situation where the omission error is higher than the commission error. Despite the small differences in metrics between the methods, the proposed TANet still demonstrates a slight advantage in overall metrics and omission errors. In addition, after adding the CFD loss, the omission error further decreases. This further validates the strong performance of our proposed network in cloud detection tasks and demonstrates that the CFD loss effectively assists the network in mitigating the issue of cloud omission.

2) Qualitative Results: In the previous analysis, we highlighted that the high similarity between the original data resulted in very close evaluation metrics for different networks on the Sentinel-2 cloud detection dataset. To visually compare the detection results of the TANet with various cloud detection methods, as well as their performance differences across different underlying surface scenarios, this section presents representative results for each type of underlying surface in the Landsat-8 cloud cover assessment dataset, as shown in Fig. 8. The first column displays the original RGB images and the true cloud masks. The subsequent columns show the error distribution maps and mask results predicted by each method. In the error distribution maps, the green color denotes the missed detection, which are cloud pixels misclassified as clear, and the red color denotes false detection, which are clear pixels that are wrongly classified as the cloud. Upon observation, we will find that cloudy areas exhibit significantly different characteristics due to variations in cloud thickness. Most methods demonstrate relatively correct predictions in regions with thick clouds, but the prediction results in the thin

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TABLE II

QUANTITATIVE RESULTS COMPARISON BETWEEN OUR METHOD AND DIFFERENT METHODS ON LANDSAT-8 CLOUD COVER ASSESSMENT VALIDATION DATA. (THE BEST PERFORMANCE IS HIGHLIGHTED IN BOLD. THE SECOND BEST PERFORMANCE IS EMPHASIZED IN UNDERLINE)

Method	Overall Accuracy (%)	Kappa Statistic	Commission Error (%)	Omission Error (%)	OE_{thick} (%)	OE_{thin} (%)
Unet	85.35	0.6621	4.623	32.77	1.011	31.75
DeeplabV3	85.31	0.6631	**4.219**	33.17	0.960	32.21
MFCNN	84.08	0.6380	4.655	35.13	1.086	34.08
CDnetV1	87.41	0.7451	4.434	26.67	0.967	25.73
CDnetV2	87.35	0.6972	5.444	27.83	0.860	26.97
MSCFF	88.99	0.7236	6.853	21.68	0.852	20.81
Boundary Nets	86.90	0.6934	4.808	29.17	1.057	28.14
TANet (w/o CFD loss)	89.71	0.7392	6.773	19.68	0.925	18.80
TANet (w/ CFD loss)	**92.26**	**0.7989**	5.822	**12.46**	**0.489**	**11.98**

TABLE III

QUANTITATIVE RESULTS COMPARISON BETWEEN OUR METHOD AND DIFFERENT METHODS ON THE SENTINEL-2 CLOUD MASK CATALOGUE DATASET

Method	Overall Accuracy (%)	Kappa Statistic	Commission Error (%)	Omission Error (%)
Unet	93.45	0.8682	5.480	7.778
DeeplabV3	93.35	0.8664	4.845	8.661
MFCNN	93.06	0.8840	4.298	9.775
CDnetV1	93.57	0.8664	4.787	8.270
CDnetV2	94.23	0.8708	3.599	8.166
MSCFF	93.99	0.8793	4.335	7.877
Boundary Nets	94.48	0.8896	**3.014**	8.194
TANet (w/o CFD loss)	**94.66**	**0.8927**	3.836	7.026
TANet (w/ CFD loss)	94.53	0.8910	4.561	**6.411**

cloud regions vary among methods, with commonly significant omissions. In comparison, our method shows more accurate detection of thin cloud regions, producing better results.

Thin clouds are sometimes found at the edge of thick clouds, such as (a) in Fig. 8. Even when some thin cloud pixels are close to thick clouds, most cloud detection methods struggle to detect thin clouds well. This is because thin clouds exhibit greater differences from thick clouds, and there are also nearby noncloud areas with more similar features, leading to errors in detecting the boundaries of cloud regions. TANet, however, can capture the subtle features of the easily confused thin cloud areas and then obtain more precise cloud regions and boundaries. In Fig. 8(b), thin and thick clouds appear together, but at a distance, separated by large areas of clear ground. Much of the area of isolated thin clouds is ignored in the results of other methods, whereas our method can successfully capture them. In addition, there are some thin clouds distributed in the image in isolation, occupying a significant portion or even all of the image, as illustrated in Fig. 8(c). In these cases, the entire image lacks distinctly visible regions with thick clouds, and the semitransparent thin clouds exhibit features similar to the underlying surface. The overall image features are relatively homogeneous. Most cloud detection methods in comparison often miss the majority of thin clouds, while our method can sensitively perceive the existence of potential thin clouds. TANet trained with the additional CFD loss provides even more precise results. This corroborates the effective collaboration between our proposed TANet and CFD loss, demonstrating their positive contributions to cloud detection in scenarios characterized by the coexistence of thin and thick clouds, as well as in scenes where exclusively thin clouds are prevalent.

Moreover, cloud detection in scenes with snow and ice poses a significant challenge due to the spectral reflectance characteristics of snow/ice and clouds being very similar in the visible light spectrum [64], [65]. For example, in Fig. 8(d), the overall image exhibits high brightness, and the contrast between light and dark areas is not necessarily due to the presence or absence of clouds but is rather influenced by the thickness of the cloud. Most of the other methods suffer from interference, recognize only the brighter parts as clouds, and miss the darker regions as cloud-free. Due to our method's comprehensive consideration of not only the commonalities among different thickness cloud regions but also the distinctiveness between thin clouds and extremely similar underlying surfaces, our approach resists the aforementioned negative impacts. Consequently, the proposed method maintains strong robustness and accuracy in challenging subsurface scenarios.

C. Ablation Study

In this section, to validate the effectiveness of the color prior guidance module, GIAM, and CFD loss respectively, ablation experiments were conducted on the baseline network. The experiments are still based on the reconstructed Landsat dataset. Initially, the baseline RS-Net [30] was used for training and prediction, and then, each module was added to the framework individually. The corresponding prediction metrics are compared separately to assess the effectiveness of each strategy. Table IV presents the quantitative evaluation results of the ablation experiments. The results indicate that each module independently and effectively improves prediction accuracy, primarily reducing the omission error, which highlights the unique contributions of each strategy to thin cloud detection. The complete method achieved the highest detection accuracy,

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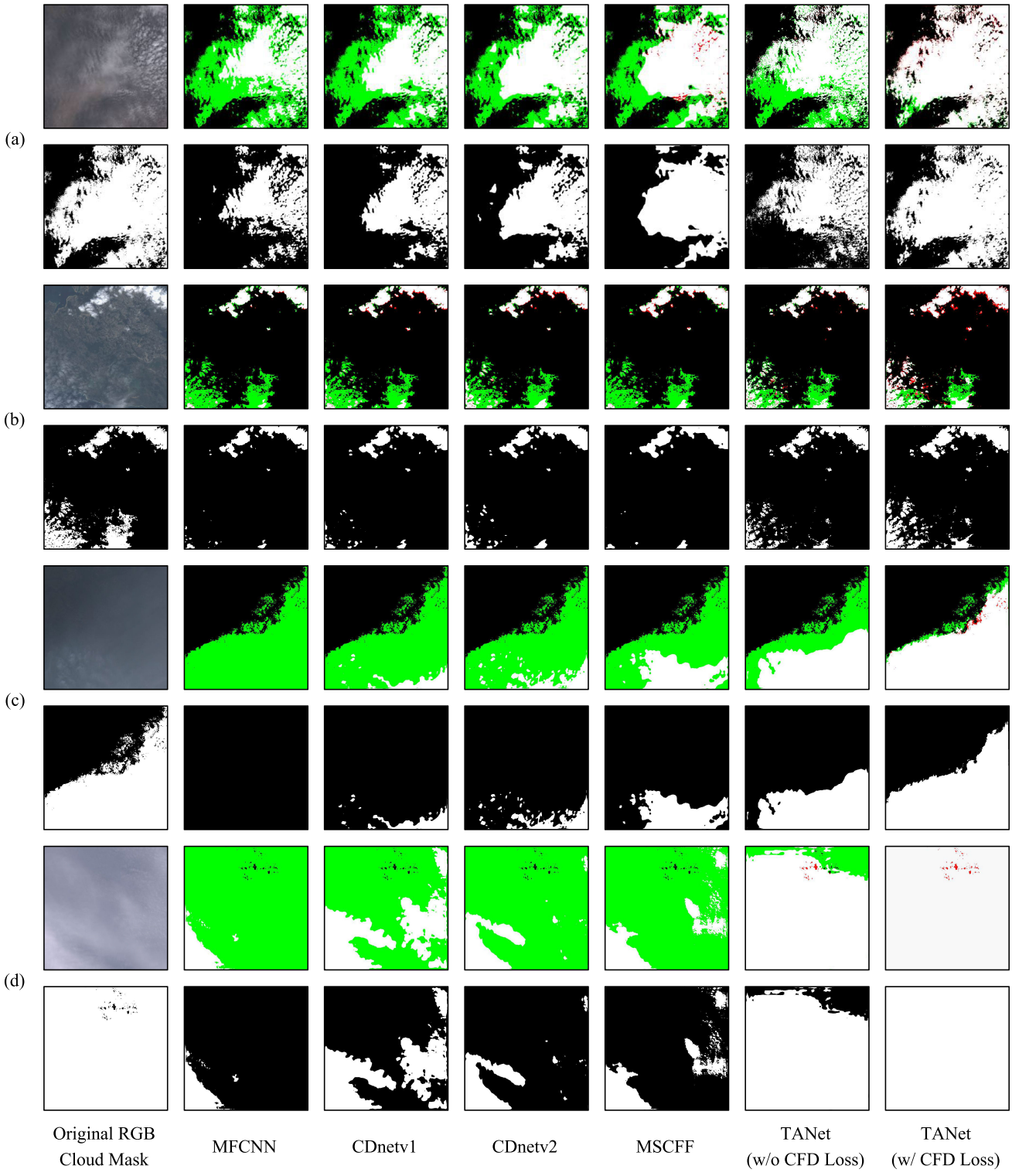


Fig. 8. Visual comparisons of the detection results in various scenes. The red part is false detection, and the green part is missed detection. (a) Shrubland. (b) Water. (c) Wetlands. (d) Snow/ice.

increased by 6.65% compared to the baseline network, and had the lowest omission error, significantly reducing it by 19.67%.

Through the partial visual results presented in Fig. 9, we observe that each module can to some extent alleviate the issue of thin cloud omissions. However, the effectiveness of each module varies slightly depending on different thin cloud scenarios. Taking the shrubland land cover type in Fig. 9(a) as an example, where thin clouds are predominantly isolated and located at a distance from thick clouds, the network with the

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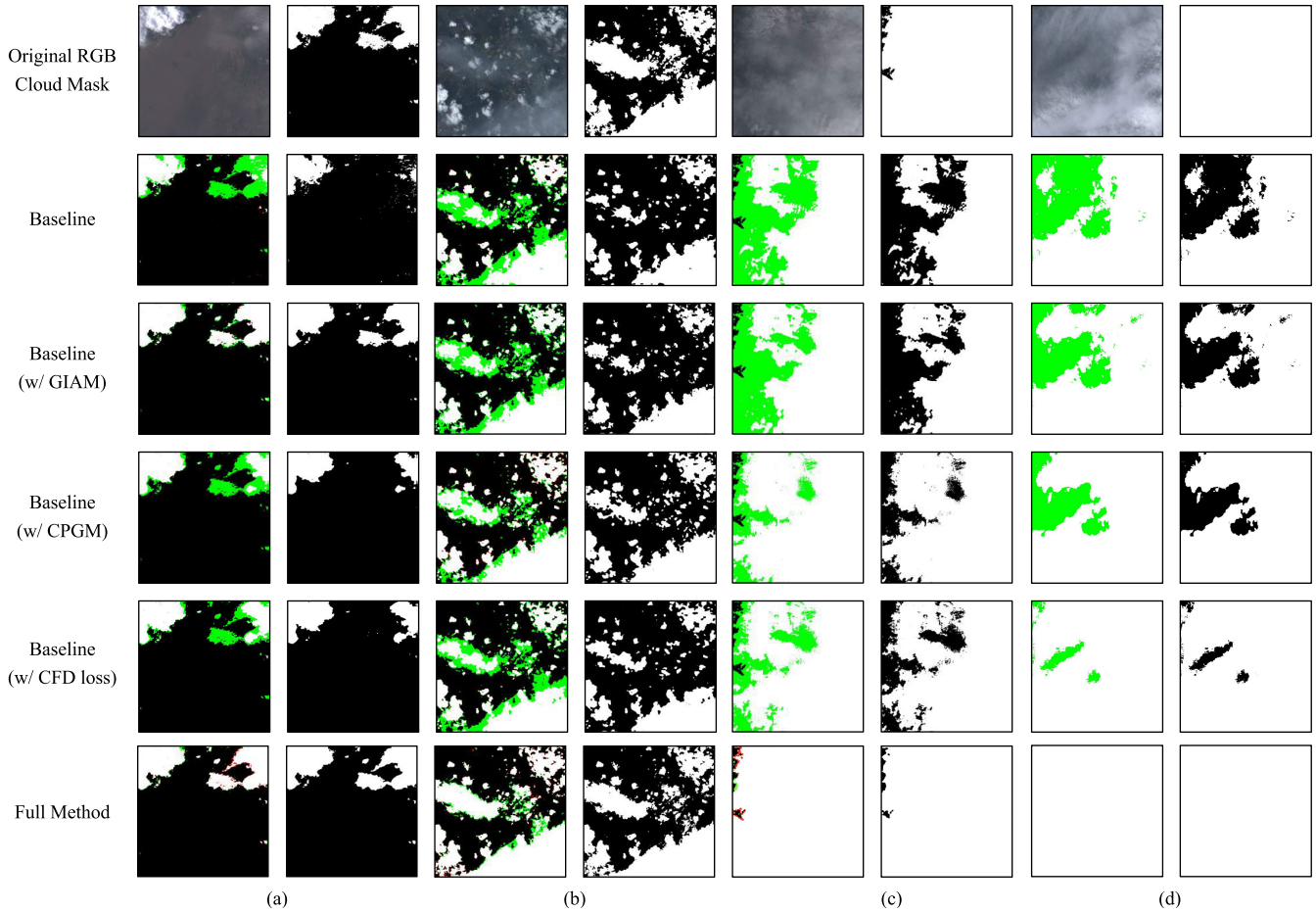


Fig. 9. Visual results of ablation study for each module. The left column is the error distribution map, the red part is false detection, and the green part is missed detection. The right column is the prediction results. (a) Shrubland. (b) Urban. (c) Forest. (d) Grass/crops.

TABLE IV

QUANTITATIVE RESULTS OF ABLATION STUDY FOR EACH OF THE PROPOSED STRATEGIES. (THE VALUES IN BRACKETS ARE THE CHANGE VALUES OF CORRESPONDING EVALUATION METRIC)

Method	Overall Accurancy (%)	Kappa Statistic	Commission Error (%)	Omission Error (%)
Baseline	85.61	0.6661	4.818	32.13
Baseline (w/ GIAM)	87.34(+1.73%)	0.6921	5.987	27.51(-4.62%)
Baseline (w/ CPGM)	88.18(+2.57%)	0.7242	**3.404**	27.85(-4.29%)
Baseline (w/ CFD loss)	87.31(+1.70%)	0.6996	4.760	28.62(-3.51%)
Full Method	**92.26(+6.65%)**	**0.7989**	5.822	**12.46(-19.67%)**

added GIAM demonstrates the most accurate detection for this specific scenario, outperforming other modules. In Fig. 9(b), more complex situations where thin and thick clouds coexist, the underlying surface in the thin cloud region is clear, and the CPGM and CFD loss show improvements in results superior to GIAM. This phenomenon may be attributed to the fact that when thin and thick clouds overlap, the assistance provided by global information is relatively limited, while extracting distinctive details between pixels of different classes and effectively suppressing underlying surface information become more crucial. Even in scenarios where thin and thick clouds almost entirely cover the entire image, as shown in (c) and (d) in Fig. 9, all three modules remain effective in solving the miss detection problem of thin clouds, and their combination achieves the best results. These examples provide additional evidence supporting the validity of our analysis of different aspects of thin clouds and confirm the rationality and effectiveness of all modules.

D. Evaluation of CFD Loss

The CFD loss can be integrated into the training process of any existing cloud detection network. To justify its effectiveness, we conducted ablation studies on the CFD loss within the above other algorithms using Landsat-8 cloud cover assessment validation data. In the experiments, other settings remained constant, and only the weight (λ) and margin of the CFD loss were manually adjusted based on multiple experimental results until optimal performance was achieved. It can be seen that in Table V, after incorporating the CFD loss in the training process, the omission errors in the outcomes of various networks were decreased, resulting in improved accuracy.

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TABLE V

QUANTITATIVE RESULTS OF ABLATION STUDY FOR THE CFD LOSS. (THE TWO ADJACENT ROWS WITHIN THE SOLID LINE INDICATE THE TRAINING RESULTS OBTAINED BEFORE AND AFTER THE USE OF CFD LOSS)

Method	Overall Accurancy (%)	Kappa Statistic	Commission Error (%)	Omission Error (%)
Unet	85.35	0.6661	4.818	32.13
Unet (w/ CFD loss)	86.81 (+1.46)	0.6914	4.447	29.96 (-2.17)
DeeplabV3	85.31	0.6631	4.219	33.17
DeeplabV3 (w/ CFD loss)	87.36 (+2.00)	0.6997	4.944	28.34 (-4.83)
MFCNN	84.08	0.6380	4.655	35.13
MFCNN(w/ CFD loss)	85.81 (+1.73)	0.6751	3.699	32.56 (-2.57)
CDnetV2	87.35	0.6972	5.444	27.83
CDnetV2 (w/ CFD loss)	88.66 (+1.31)	0.7239	5.434	24.67 (-3.16)
MSCFF	88.99	0.7236	6.853	21.68
MSCFF(w/ CFD loss)	90.78 (+1.79)	0.8080	6.196	17.52 (-4.16)

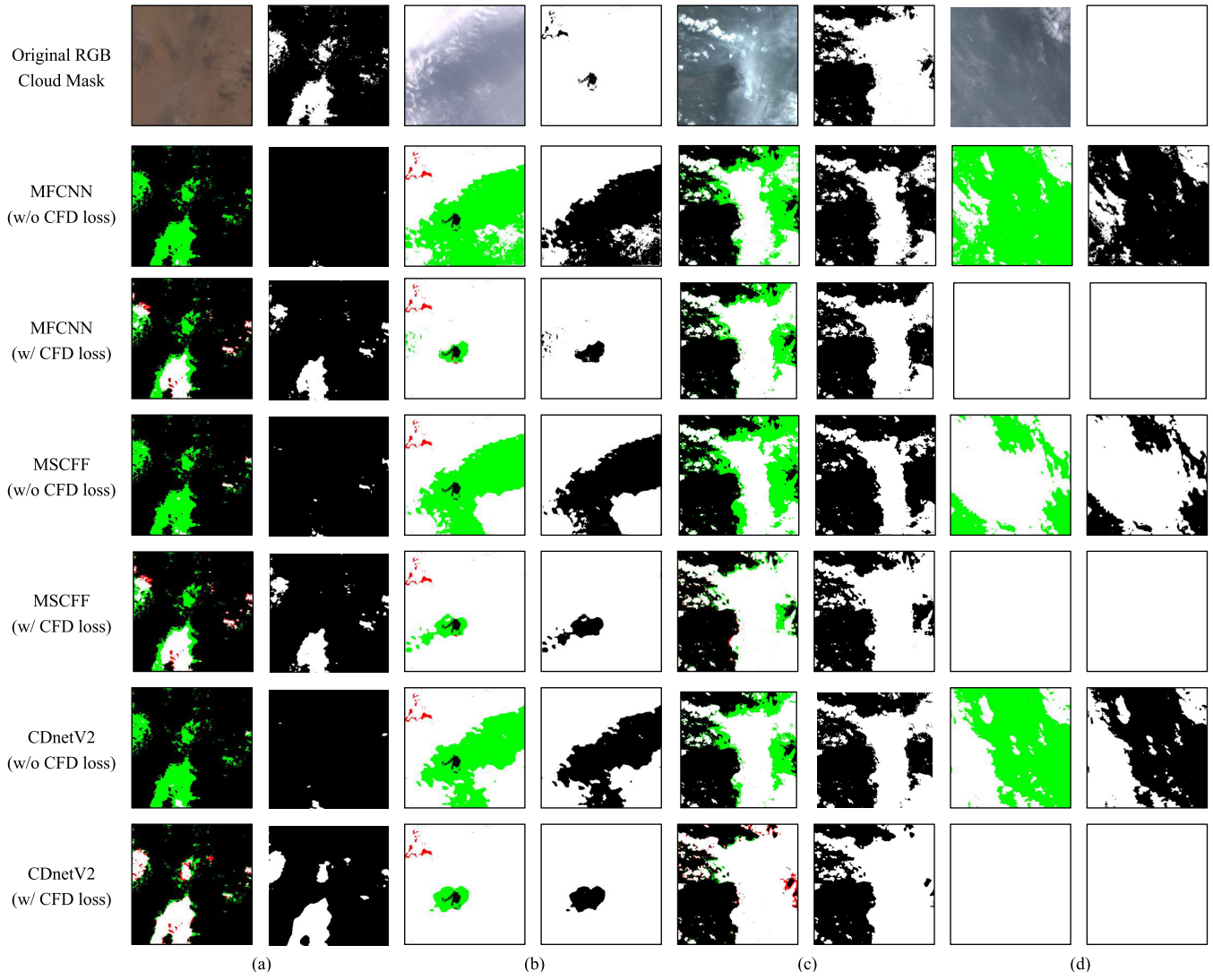


Fig. 10. Visual results of ablation study for CFD loss. (a) Barren. (b) Snow/ice. (c) Grass/crops. (d) Wetlands. From top to bottom is a comparison of the prediction results of different cloud detection networks before and after training with CFD loss. The left column is the error distribution map, and the right column is the prediction results.

Fig. 10 compares some cloud detection networks under different subsurface conditions, demonstrating the results before and after incorporating the CFD loss for contrast. From Fig. 10(a), it can be observed that some thin clouds are challenging to distinguish in the image, and most methods struggle to accurately detect their presence. Training the network with the CFD loss proves effective in enhancing the exact identification of thin clouds. Cloud detection over snow/ice surfaces is a difficult task, as illustrated in Fig. 10(b), where the overall image exhibits high brightness and differences

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in cloud thickness lead to variations in regional brightness. This may cause most methods to incorrectly classify darker regions as cloud-free regions. The CFD loss, by forcing the network to learn the differences between different categories and similarities among the same categories, effectively corrects the error detection of this part by each network in this area. In some scenarios, thin and thick clouds may appear mixed. In certain situations, the difference between thin clouds and thick clouds is significant, while the difference between thin clouds and underlying surfaces is smaller, as shown in Fig. 10(c). Sometimes thin clouds take up most of the image and are missed by varying degrees of large areas, and only a small portion of the brighter thicker clouds can be detected, i.e., Fig. 10(d). Regardless of the scenario, networks trained with the additional CFD loss can still reliably detect thin clouds and yield more reliable overall results. The quantitative and qualitative results indicate that our CFD loss effectively enhances the network's perception of thin clouds.

IV. Conclusion

Aiming at the challenge of overlooking thin clouds in detection, we introduce a TANet and a CFD loss. Through the introduced CPGM, GIAM, and CFD loss, the network becomes more focused on the region with thin clouds, acquiring global information and differentiated features that facilitate the identification of easily confused cloud samples. The CPGM utilizes robust dehaze color priors to guide the network's focus toward thin cloud regions. The GIAM establishes nonlocal structures to capture long-distance dependence information between pixels, which helps counteract the adverse effects of local subsurface features in thin cloud detection. By incorporating the CFD loss in training the proposed network, the capability of modeling differentiated details between thin cloud and cloud-free regions is enhanced. The experimental results on the Landsat-8and Sentinel-2 cloud detection dataset indicate that compared to other cloud detection methods, TANet achieves the lowest omission error and the highest overall accuracy. This confirms its excellent performance in precise identification of thin clouds, so as to obtain more reliable overall detection results. Simultaneously, we conducted ablation experiments on each module, confirming that each strategy can independently improve thin cloud detection and complement each other, playing a positive role in achieving better cloud detection. In addition, the proposed CFD loss can be well integrated into the existing cloud detection network to guide the training and effectively strengthen the network's ability to accurately identify thin clouds.

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